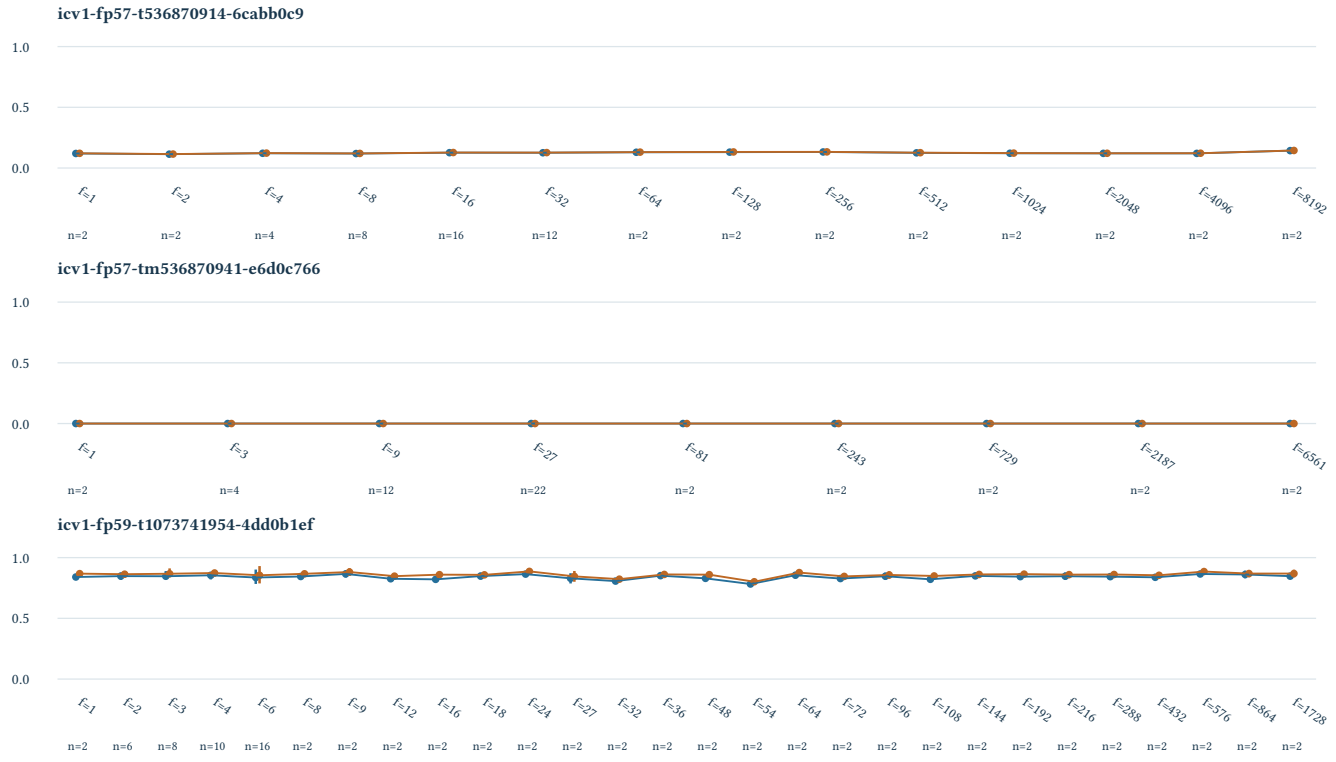
Volcano IC: all strata, 99-vertex checkpoint

Run 20261017T0337Z. Counted diagnostic; no admitted runtime or speedup claim.

All 51 conductor strata: measured size-64 MITM yields

Blue: relations per source trial. Orange: relations per actual oracle call. Points: native mean; bars: observed min–max.

Conductor order is ordinal on each x-axis; n is measured target attempts, not independent vertices or a confidence interval.



Run 20261017T0337Z: 99 audited vertices / 24,823 planned. All strata covered by samples; entire-graph distributions remain incomplete.

Graph key	Audited vertices	Planned vertices	Audited strata	Planned strata	Largest End(E) conductor
mixed	44	10070	28	28	1728
three	25	6561	9	9	6561
two	30	8192	14	14	8192

Observed: 1,584 measured attempts: 1,065 verified recoveries, 422 exhausted budgets, 97 watchdog timeouts. All 51 conductor strata have measured vertices; only 99 of 24,823 vertices have been audited. Stratum coverage is not complete graph coverage. Maximal sampled conductors are 8192, 6561 and 1728.

Integrity: every standard audit receipt passes; successful/exhausted deterministic reports replayed when collected. The stopped checkpoint inventory rechecks every receipt/file hash, all 1,584 record seals, exact replay populations and ledger/queue identities. It is a receipt-binding check, not a fresh or independent solver replay. Exact inventory and raw frozen files accompany this report.

Yield definitions and complete stratum tables

Correction to earlier wording: the earlier reports called 301,715 source trials “attempted decompositions.” Trials include duplicate queries skipped before the oracle. Their recorded values and historical PDFs are unchanged. The new summary retains both denominators: 2,223,204 source trials and 2,075,145 actual decomposition calls, producing 57,176 relations across 990 IC reports. For these guarded classic-walk arms, calls equal trials minus repeated queries skipped. Identity skips occur before trial increment. All 594 two-summand MITM records independently agree with their lookup counters. A zero/unknown call denominator has no yield; timeout counters are never assumed zero.

The legacy field relation-yield remains a per-trial alias. Summary schema v2 adds explicit trial-relation-yield and oracle-call-relation-yield, numerator, denominator, sample counts and definitions. Global totals across distinct subgroups/bases are descriptive accounting, not a common success probability.

The tables show only size-64 projected MITM, with every conductor stratum. GAE values are lower bounds with unpriced native work. Exhausted rows are attempt costs, not successful DLP costs. Means use present observations and the explicit n; multiple vertices per stratum are sampled, not a census. Native CSV/JSON contain every arm, phase, status, range, matrix rank/dependency, memory and raw elapsed observation. Exact generator representations remain explicit and unregistered where noted; no new curve-name alias is inferred.

icv1-fp57-t536870914-6cabb0c9

Conductor	Status	n	GAE min	GAE mean	GAE max	Mean/trial	Mean/call
1	verified	2	19,530	19,970.0	20,410	0.119769	0.121079
2	verified	2	19,505	19,546.5	19,588	0.114059	0.114919
4	verified	4	19,466	20,048.5	20,714	0.121453	0.122819
8	verified	8	19,467	19,888.2	20,857	0.119064	0.120233
16	verified	16	19,210	19,796.4	20,524	0.126336	0.127545
32	verified	12	19,434	19,969.1	20,954	0.125741	0.127051
64	verified	2	19,797	19,997.0	20,197	0.129674	0.130887
128	verified	2	19,324	19,538.5	19,753	0.130729	0.132345
256	verified	2	19,260	20,026.5	20,793	0.132167	0.132942
512	verified	2	19,514	19,930.5	20,347	0.125041	0.126247
1024	verified	2	19,921	20,062.0	20,203	0.121524	0.122809
2048	verified	2	19,076	19,585.0	20,094	0.120734	0.121480
4096	verified	2	19,609	20,067.0	20,525	0.120562	0.121898
8192	verified	2	19,369	19,846.5	20,324	0.143079	0.144412

icv1-fp57-tm536870941-e6d0c766

Conductor	Status	n	GAE min	GAE mean	GAE max	Mean/trial	Mean/call
1	exhausted	2	20,239	20,252.0	20,265	0.000000	0.000000
3	exhausted	4	20,245	20,270.0	20,298	0.000000	0.000000
9	exhausted	12	20,240	20,273.7	20,327	0.000000	0.000000
27	exhausted	22	20,226	20,265.6	20,299	0.000000	0.000000
81	exhausted	2	20,302	20,319.0	20,336	0.000000	0.000000
243	exhausted	2	20,284	20,299.5	20,315	0.000000	0.000000
729	exhausted	2	20,278	20,283.0	20,288	0.000000	0.000000
2187	exhausted	2	20,253	20,270.0	20,287	0.000000	0.000000
6561	exhausted	2	20,283	20,286.0	20,289	0.000000	0.000000

icv1-fp59-t1073741954-4dd0b1ef

Conductor	Status	n	GAE min	GAE mean	GAE max	Mean/trial	Mean/call
1	verified	2	20,463	20,467.5	20,472	0.840387	0.868095
2	verified	6	20,012	20,126.2	20,241	0.847810	0.863518
3	verified	8	20,024	20,270.8	20,595	0.846633	0.867332
4	verified	10	19,899	20,159.5	20,527	0.854733	0.873263
6	verified	16	19,900	20,144.6	20,550	0.836148	0.854720
8	verified	2	20,050	20,250.0	20,450	0.844634	0.866823
9	verified	2	20,080	20,115.0	20,150	0.865867	0.881503
12	verified	2	19,813	19,988.0	20,163	0.826053	0.847380
16	verified	2	20,142	20,300.0	20,458	0.821076	0.859763
18	verified	2	20,008	20,008.0	20,008	0.848025	0.858225
24	verified	2	20,086	20,297.0	20,508	0.864248	0.886644
27	verified	2	19,951	19,992.0	20,033	0.829576	0.845611
32	verified	2	19,880	20,051.5	20,223	0.806809	0.822622
36	verified	2	19,875	19,990.0	20,105	0.851496	0.861641
48	verified	2	20,241	20,297.5	20,354	0.829370	0.859703
54	verified	2	20,100	20,328.5	20,557	0.782841	0.801320
64	verified	2	19,996	20,194.5	20,393	0.855256	0.876868
72	verified	2	19,964	20,203.5	20,443	0.827778	0.845433
96	verified	2	19,919	19,929.5	19,940	0.846732	0.857399
108	verified	2	20,273	20,492.0	20,711	0.821501	0.849152
144	verified	2	19,922	20,056.0	20,190	0.849713	0.861567
192	verified	2	20,115	20,186.5	20,258	0.843428	0.864101
216	verified	2	20,022	20,022.5	20,023	0.846824	0.859916
288	verified	2	20,118	20,180.0	20,242	0.843369	0.861071
432	verified	2	19,933	20,043.0	20,153	0.838037	0.854658
576	verified	2	20,153	20,259.0	20,365	0.865327	0.885298
864	verified	2	19,900	19,942.0	19,984	0.861202	0.868679
1728	verified	2	20,215	20,372.5	20,530	0.847351	0.868859

Protocol and provenance: projected bases 16/64 with MITM/subtraction, unchanged raw-base MITM, rho.negation, textbook BSGS and rho A/A. Two public targets per vertex, one alternating round, zero warmups, target seed 20261017, order seed 1, 4,096 source trials and a 3-second child watchdog. Actual largest prime subgroups are retained, including the 45-bit subgroup. Cold solver setup, projection and verification are charged. All original measured sessions remain byte-frozen. Source commits 4eecf6b53 and ee54ad603; complete stopped campaign, native source, queue, ledger, receipts and frozen runner are in the archive.

Host and limits: Intel Xeon Platinum 8573C, Linux x86-64, four-CPU quota, 16 GiB RAM. Dev-profile ecbench hash 1869ca2e7be271794df8128c08a6dca0201038e2c81a3e0ce6434f7ec0d8379d. All wall observations are L0/debug and unqualified. No builds/test suites ran during the resumed measured block; it was stopped before the release build. Raw process time includes orchestration; no optimized speedup is inferred.

The full million-trial/eight-target/five-round/four-size matrix, original A–E full graph request, all-vertex completion, exact representation registry, transported-target benchmarks, composite PH, actual finite-field IC and independent implementation replay remain open. Exact embedding degrees 30,875; 3,953,777,913,734; and 2,126 exceed the implemented MOV teaching backend. This is an implementation gap, not proof that mathematical reductions do not exist. A separately named release-profile survey must pass matched count checks and audits before launch. Canonical fastest-method claims are unchanged.